VII SEMESTER (PROFESSIONAL ELECTIVE-IV)	L	T	P	C
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20IT7E05 - BIG DATA ANALYTICS				

COURSE OUTCOMES

After the completion of the course the students are able to

CO1: Explain the basic concepts of Big Data and Big Data Analytics

CO2: Understand the HDFS architecture

CO3: Design and develop the Map Reduce application

CO4: Identify and understand various Hadoop Ecosystem technologies

CO5: Apply the Advanced Analytical methods for classification, clustering and Text Analysis

UNIT -I: Introduction

Introduction to Big Data: Types of Digital Data-Characteristics of Data – Evolution of Big Data – Definition of Big Data – Challenges with Big Data – 3Vs of Big Data – Non Definitional traits of Big Data – Business Intelligence vs. Big Data – Data warehouse and Hadoop environment – Coexistence.

Big Data Analytics: Classification of analytics – Data Science – Terminologies in Big Data – CAP Theorem

UNIT -II: Hadoop Distributed File System

Introduction to Hadoop, frame work, features, advantages, HDFS concepts, HDFS architecture, Command Line Interface, hdfs commands, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives,

UNIT -III: Map Reduce

Understanding Hadoop API for MapReduce Framework, features, Techniques to Optimize MapReduce Jobs; Mapper – Reducer – Combiner – Partitioner – Searching – Sorting – Compression, I/O formats, Uses of MapReduce, Ingestion layer in big data stack

UNIT -IV: Hadoop Ecosystem

Serialization: AVRO, Co-ordination: Zookeeper. Databases: HBase Concepts, Clients, Example, HBase Versus RDBMS, Hive-architecture, data types, file formats, HQL. Scripting language: Pig-features, anatomy, pig on Hadoop, pig latin over view, data types, running pig, execution modes of pig, Streaming: Flink, Storm

UNIT -V: Advanced Analytics

Clustering: k-means, k-mode algorithm, Association Rules: Apriori Algorithm, Evaluation of Candidate Rules, Applications of Association Rules, Classification: Decision Trees, Naïve Bayes. Time Series Analysis: Overview of Time Series Analysis, ARIMA Model.

Text Analysis: Regular expressions, Collecting Raw Text, Representing Text (bag of words), Term Frequency—Inverse Document Frequency (TFIDF)

Text Books:

1. Seema Acharya, Subhashini Chellappan, “Big Data and Analytics”, Wiley Publication, 2015.
2. Tom White, “Hadoop: The Definitive Guide”, O’Reilly, 4th Edition, 2015.
3. Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data by EMC Education Services- Wiley

Reference Books:

1. Hadoop in Practice by Alex Holmes, MANNING Publ.
2. Hadoop Map Reduce Cookbook, Srinath Perera, Thilina Gunarathne.
3. Big Data Analytics with R and Hadoop-Vignesh Prajapati.

Software Links:

E-Resources:

1. Hadoop:http://hadoop.apache.org/
2. Hive: https://cwiki.apache.org/confluence/display/Hive/Home
3. Piglatin: http://pig.apache.org/docs/r0.7.0/tutorial.html